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[1]

$$(1) \int_0^{2\pi} \cos^{2n} \theta d\theta, n \in \mathbb{Z}$$

$$z = e^{i\theta}, dz = i \cdot e^{i\theta} d\theta, \cos \theta = \frac{z + z^{-1}}{2}$$

$$\text{原式} = \int_{|z|=1} \left(\frac{z + z^{-1}}{2} \right)^{2n} \frac{dz}{iz}$$

$$= \frac{1}{i} \int_{|z|=1} \left(\frac{z^2 + 1}{2z} \right)^{2n} \frac{1}{z} dz$$

$$= \frac{1}{i} \frac{1}{2^{2n}} \int_{|z|=1} \frac{(z^2 + 1)^{2n}}{z^{2n+1}} dz$$

$$\text{对于} \int_{|z|=1} \frac{(z^2 + 1)^{2n}}{z^{2n+1}} dz, z=0 \text{ 为其 } 2n+1 \text{ 阶极点,}$$

$$\text{即, 原式} = 2\pi i \cdot \frac{1}{(2n)!} \frac{d^{2n}}{dz^{2n}} [(z-0)^{2n+1} f(z)] \Big|_{z=0}$$

$$= 2\pi i \cdot \frac{1}{(2n)!} \left[\frac{d^{2n}}{dz^{2n}} [(z^2 + 1)^{2n}] \right] \Big|_{z=0}$$

$$\text{对于} \frac{d^{2n}}{dz^{2n}} (z^2 + 1)^{2n}$$

$$\text{对于} (z^2 + 1)^{2n} = \sum_{k=0}^{2n} C_{2n}^k z^{2(2n-k)}, k$$

$$\text{其中 } k < n \Rightarrow \frac{d^{2n}}{dz^{2n}} () = 0$$

$$k = n \Rightarrow \frac{d^{2n}}{dz^{2n}} () = C_{2n}^n \cdot (2n)!$$

$$k > n \Rightarrow \frac{d^{2n}}{dz^{2n}} () = \sim z^2 \Big|_{z=0} = 0$$

$$\Rightarrow \frac{d^{2n}}{dz^{2n}} (z^2+1)^{2n} = C_{2n}^n (2n)! = (2n)! \cdot \frac{(2n)!}{n! n!}$$

$$\Rightarrow \int_{|z|=1} \frac{(z^2+1)^{2n}}{z^{2n+1}} dz = 2\pi i \cdot \frac{(2n)!}{n! n!}$$

$$\Rightarrow \int_0^{2\pi} \cos^{2n} \theta d\theta = \frac{1}{i} \cdot \frac{1}{2^{2n}} 2\pi i \cdot \frac{(2n)!}{n! n!}$$

$$= \frac{\pi}{2^{2n-1}} \cdot \frac{(2n)!}{n! n!}$$

$$(2) \int_0^{2\pi} \frac{dx}{(a+b\cos x)^2}, \quad a > b > 0$$

$$z = e^{ix}, \quad dz = iz dx,$$

$$\Rightarrow \int_{|z|=1} \frac{1}{(a+b\frac{z+z^{-1}}{2})^2} \cdot \frac{1}{iz} dz$$

$$= \int_{|z|=1} \frac{z}{(a z + \frac{b}{2} z^2 + \frac{b}{2})^2} \cdot \frac{1}{i} dz$$

$$= \frac{1}{i} \int_{|z|=1} \frac{4z}{(bz^2 + 2az + b)^2} dz$$

$$\text{对于 } |z|=1 \quad \frac{4z}{(bz^2 + 2az + b)^2} dz$$

$$bz^2 + 2az + b \Rightarrow \Delta = 4a^2 - 4b^2$$

$$z_{1/2} = \frac{-2a \pm 2\sqrt{a^2 - b^2}}{2b} = \frac{-a \pm \sqrt{a^2 - b^2}}{b}$$

对于 $|z_2| > 1$, 可以舍去

$$\int_0^{2\pi} \frac{dx}{(a + b \cos x)^2} = 2\pi \left\{ \frac{d}{dz} \left[(z - z_1)^2 \frac{4z}{b^2(z - z_1)^2(z - z_2)^2} \right] \right\} \Big|_{z=z_1}$$

$$= 2\pi \left\{ \frac{d}{dz} \left[\frac{4z}{b^2(z - z_2)^2} \right] \right\} \Big|_{z=z_1}$$

$$= 2\pi \left[\frac{1}{b^4(z - z_2)^4} (4b^2(z - z_2)^2 - 4z \cdot 2b^2(z - z_2)) \right] \Big|_{z=z_1}$$

$$\Rightarrow z_1 - z_2 = \frac{-a + \sqrt{a^2 - b^2} + a + \sqrt{a^2 - b^2}}{b} = \frac{2\sqrt{a^2 - b^2}}{b}$$

$$\text{原式} = 2\pi \left[\frac{1}{(a^2 - b^2)^2} (1b(a^2 - b^2) - 8 \frac{-a + \sqrt{a^2 - b^2}}{b} b^2 \frac{2\sqrt{a^2 - b^2}}{b}) \right]$$

$$= 2\pi \left[\frac{1}{(a^2 - b^2)^2} (a^2 - b^2 - (-a + \sqrt{a^2 - b^2})(\sqrt{a^2 - b^2})) \right]$$

$$= 2\pi \left[\frac{1}{(a^2-b^2)^{\frac{3}{2}}} (\sqrt{a^2-b^2} + a - \sqrt{a^2-b^2}) \right]$$

$$= 2\pi \frac{a}{(a^2-b^2)^{\frac{3}{2}}}$$

$$= \frac{2a\pi}{(a^2-b^2)^{\frac{3}{2}}}$$

$$(3) \int_0^\pi \frac{d\theta}{1+\sin^2\theta}$$

$$\alpha = 2\theta, \quad d\alpha = 2d\theta$$

$$\Rightarrow \int_0^{2\pi} \frac{1}{2} \frac{d\alpha}{1+\sin^2\frac{\alpha}{2}}$$

$$= \int_0^{2\pi} \frac{1}{2} \frac{d\alpha}{\frac{1}{2}[3-\cos\alpha]}$$

$$= \int_0^{2\pi} \frac{d\alpha}{3-\cos\alpha}$$

$$\Rightarrow = \int_{|z|=1} \frac{1}{3 - \frac{z+z^{-1}}{2}} \cdot \frac{1}{i} dz$$

$$= \frac{1}{i} \int_{|z|=1} \frac{1}{3z - \frac{z^2+1}{2}} dz$$

$$= \frac{1}{i} \int_{|z|=1} \frac{z}{1 - (z^2 - 6z + 1)} dz$$

$$= -\frac{z}{i} \int_{|z|=1} \frac{1}{z^2 - 6z + 1} dz$$

对于 $\int_{|z|=1} \frac{1}{z^2 - 6z + 1} dz$

$$z_1 = \frac{6 + 4\sqrt{2}}{2}, \quad z_2 = \frac{6 - 4\sqrt{2}}{2} = 3 - 2\sqrt{2}$$

对于 $|z| > 1$ 无法使用 residue

$$\Rightarrow \int_{|z|=1} \frac{1}{z^2 - 6z + 1} dz$$

$$= 2\pi i \cdot \left[(1 - z - z_2) \frac{1}{(z - z_1)(z - z_2)} \right] \Big|_{z=z_2}$$

$$= 2\pi i \cdot \frac{1}{-4\sqrt{2}}$$

$$= \frac{\pi i}{-2\sqrt{2}}$$

$$\Rightarrow \int_0^\pi \frac{d\theta}{1 + \sin^2 \theta} = \frac{-2}{i} \cdot \frac{\pi i}{-2\sqrt{2}}$$

$$= \frac{\pi}{\sqrt{2}}$$

$$(4) \int_0^\pi \frac{d\theta}{(1+\sin^2\theta)^2} \quad \text{变换过程与(3)同理}$$

$$= \int_0^{2\pi} \frac{1}{2} \frac{d\alpha}{\frac{1}{4}(3-\cos\alpha)^2}$$

$$= 2 \int_0^{2\pi} \frac{d\alpha}{(3-\cos\alpha)^2}$$

$$= 2 \int_{|z|=1} \frac{1}{(3 - \frac{z+\bar{z}}{2})^2} \frac{1}{iz} dz$$

$$= \frac{2}{i} \int_{|z|=1} \frac{4z}{(6z - z^2 - 1)^2} dz$$

$$= \frac{2}{i} \int_{|z|=1} \frac{4z}{(z^2 - 6z + 1)^2} dz$$

$$z_1 = 3 + 2\sqrt{2}, \quad z_2 = 3 - 2\sqrt{2}, \quad |z_i| > 1$$

$$z_1, z_2 \Rightarrow \text{二阶极点}$$

$$= \frac{2}{i} 2\pi i \left[\frac{d}{dz} (z - z_2)^2 \frac{4z}{(z - z_1)^2 (z - z_2)^2} \right] \Big|_{z=z_2}$$

$$= 4\pi \left[\frac{d}{dz} \frac{4z}{(z - z_1)^2} \right] \Big|_{z=z_2}$$

$$= 4\pi \left[\frac{1}{(z - z_1)^3} (4(z - z_1)^2 - 4z \cdot 2(z - z_1)) \right] \Big|_{z=z_2}$$

$$z_2 - z_1 = 3 - 2\sqrt{2} - 3 - 2\sqrt{2} = -4\sqrt{2}$$

$$\text{原式} = 4\pi \left[\frac{1}{(4\sqrt{2})^4} (4 \cdot (4\sqrt{2})^2 + 8(3 - 2\sqrt{2})4\sqrt{2}) \right]$$

$$= 4\pi \left[\frac{1}{(4\sqrt{2})^3} (16\sqrt{2} + 24 - 16\sqrt{2}) \right]$$

$$= 4\pi \cdot \frac{4 \cdot 8 \cdot 3}{4 \cdot 16 \cdot 8 \cdot \sqrt{2}}$$

$$= \frac{3\pi}{4\sqrt{2}}$$

[2]

$$(1) \int_{-\infty}^{\infty} \frac{x^2}{1+x^4} dx$$

$$= \oint_C \frac{z^2}{1+z^4} dz$$

$$\Rightarrow (1-i) \int_0^R \frac{z^2}{1+z^4} dz + \int_{CR} \frac{z^2}{1+z^4} dz$$

$$= 2\pi i \operatorname{res} \frac{z^2}{1+z^4} \Big|_{z=e^{i\frac{\pi}{4}}} \cdot (1-i)$$

$$= 2\pi i \cdot (1-i) \left(\frac{z^2}{(z+e^{i\frac{\pi}{4}})(z^2+e^{i\frac{\pi}{2}})} \right) \Big|_{z=e^{i\frac{\pi}{4}}}$$

$$= 2\pi i \cdot \frac{i}{\sqrt{2}(1+i)(i+i)} \cdot (1-i)$$

$$= \frac{\pi i}{\sqrt{2}(1+i)} \cdot (1-i)$$

$$= (1-i) \cdot \frac{\pi i}{\sqrt{2}}$$

$$= \frac{\pi}{\sqrt{2}}$$

$$(2) \int_{-\infty}^{\infty} \frac{1}{(1+x^2)^{n+1}} dx, \quad n \in \mathbb{Z}$$

$$\Rightarrow \int_{-R}^R \frac{1}{(1+z^2)^{n+1}} dz + \int_R \frac{1}{(1+z^2)^{n+1}} dz$$

$$= 2\pi i \operatorname{res} \frac{1}{(1+z^2)^{n+1}} \Big|_{z=e^{i\frac{\pi}{2}}}$$

$$= 2\pi i \cdot \frac{d^n}{dz^n} (z - e^{i\frac{\pi}{2}})^{n+1} \frac{1}{(z - e^{i\frac{\pi}{2}})^{n+1} (z + e^{i\frac{\pi}{2}})^{n+1}} \Big|_{z=e^{i\frac{\pi}{2}}}$$

$$= \frac{2\pi i}{n!} \cdot \frac{d^n}{dz^n} (z + e^{i\frac{\pi}{2}})^{-(n+1)} \Big|_{z=e^{i\frac{\pi}{2}}}$$

$$= \frac{2\pi i}{n!} \cdot (-1)^n \frac{(2n)!}{n!} \left[(z + e^{i\frac{\pi}{2}})^{-(2n+1)} \right] \Big|_{z=e^{i\frac{\pi}{2}}}$$

对于 $e^{i\frac{\pi}{2}} = i$

$$\Rightarrow = 2\pi i (-1)^n \frac{(2n)!}{n!n!} (zi)^{-(2n+1)}$$

$$= 2\pi i (i)^{2n} \frac{(2n)!}{n!n!} z^{-(2n+1)} (i)^{-2n} \cdot i^{-1}$$

$$= 2\pi \cdot \frac{(2n)!}{n!n!} \frac{1}{z^{2n+1}}$$

$$= \frac{\pi}{z^{2n}} \cdot \frac{(2n)!}{n!n!}$$

(3) $\int_{-\infty}^{+\infty} \frac{x^{2m}}{1+x^{2n}} dx, n, m \in \mathbb{Z} \text{ \& } n > m$

$$= 2 \int_0^{\infty} \frac{x^{2m}}{1+x^{2n}} dx$$

$$= \frac{1}{n} \int_0^{\infty} \frac{y^{\frac{2m+1}{2n}}}{(1+y)y} dy$$

$$\Rightarrow \int_0^{\infty} \frac{y^{\frac{2m+1}{2n}}}{(1+y^2)dy} dy = \pi \csc\left(\frac{2m+1}{2n}\pi\right)$$

$$\Rightarrow \int_{-\infty}^{+\infty} \frac{x^{2m}}{1+x^{2n}} dx = \frac{\pi}{n} \csc\left(\frac{2m+1}{2n}\pi\right)$$

$$(4) \int_{-\infty}^{\infty} \frac{dx}{(1+x^2) \cosh(\frac{\pi}{2}x)}$$

$$\cosh\left(\frac{\pi}{2}z\right) = 0 \Rightarrow z = (2n+1)i, \quad n \in \mathbb{Z}$$

$$f(z) = \frac{1}{(z^2+1) \cosh(\frac{\pi}{2}z)}$$

$$\text{Res}(f, z_n) = \frac{1}{2\pi i} \int_{|z-z_n|=\rho} \frac{1}{z-z_n} \cdot \frac{\frac{\pi}{2}z - \frac{\pi}{2}z_n}{\cosh(\frac{\pi}{2}z) - \cosh(\frac{\pi}{2}z_n)} \frac{dz}{\frac{\pi}{2}(z^2+1)}$$

$$= \frac{1}{\sinh(\frac{\pi}{2}z_n)} \cdot \frac{1}{\frac{\pi}{2}(z_n^2+1)}$$

$$= \frac{1}{2\pi} \cdot \frac{(-1)^{n+1}}{n \cdot (n+1)}$$

$$h(z) = \frac{\pi}{z} \frac{z-i}{\cosh(\frac{\pi}{2}z) - \cosh(\frac{\pi}{2}i)}$$

$$h(i) = \frac{1}{\sinh(\frac{\pi}{2}i)} = -i$$

$$h'(i) = \lim_{z \rightarrow i} \frac{\frac{\pi}{2} \cosh(\frac{\pi}{2}z) - \frac{\pi}{2}(z-i) \frac{\pi}{2} \sinh(\frac{\pi}{2}z)}{[\cosh(\frac{\pi}{2}z)]^2}$$

$$= \lim_{z \rightarrow i} \frac{-\frac{\pi}{2}(z-i)}{\pi \sinh(\frac{\pi}{2}z)} \cdot \frac{\pi^2}{4}$$

$$= 0$$

$$\text{Res}(f, i) = \frac{d}{dz} \left[\frac{zh(z)}{\pi(z+i)} \right] \Big|_{z=i}$$

$$= \frac{z[h'(z)(z+i) - h(z)]}{\pi(z+i)^2} \Big|_{z=i}$$

$$= \frac{1}{2\pi i}$$

$$\Rightarrow \sum_{n=0}^{\infty} \text{Res}(f, z_n) = \frac{1}{2\pi i} + \sum_{n=1}^{\infty} \frac{i(-1)^{n+1}}{2\pi n(n+1)}$$

$$= \frac{1}{2\pi i} + \frac{1}{2\pi i} (-1 + 2 \ln^2)$$

$$= \frac{\ln^2}{\pi i}$$

$$\int_{-\infty}^{+\infty} f(x) dx = \lim_{N \rightarrow \infty} \oint_{CN} f(z) dz$$

$$= \lim_{N \rightarrow \infty} 2\pi i \sum_{n=0}^{2N-1} \text{Res}(f, z_n)$$

$$= 2\pi i \sum_{n=0}^{\infty} \text{Res}(f, z_n)$$

$$= 2\pi i \cdot \frac{\ln^2}{\pi i}$$

$$= 2\ln^2$$

[3]

$$(1) \int_0^{\infty} \frac{\cos x}{1+x^4} dx$$

$$= \frac{1}{2} \int_{-\infty}^{+\infty} \frac{\cos z}{1+z^4} dz$$

$$\cos z = e^{iz}, \quad f(z) = \frac{\cos z}{1+z^4}$$

$$\Rightarrow \int_R f(z) dz + \int_{CR} f(z) dz$$

$$= 2\pi i \cdot \text{Res}[f, z_1) + (f, z_2)]$$

$$\begin{aligned}\operatorname{Res}(f, z_1) &= \lim_{z \rightarrow z_1} \frac{z - z_1}{1 + z^4} e^{iz} \\ &= -\frac{(1+i)}{4\sqrt{2}} e^{\frac{(i-1)}{\sqrt{2}}}\end{aligned}$$

$$\operatorname{Res}(f, z_2) = -\frac{(1-i)}{4\sqrt{2}} e^{\frac{-i-1}{\sqrt{2}}}$$

$$2\pi i \cdot (\operatorname{Res}(f, z_1) + \operatorname{Res}(f, z_2))$$

$$= 2\pi i \cdot \frac{-1}{4\sqrt{2}} e^{-\frac{1}{\sqrt{2}}} \cdot 2i \left(\cos \frac{1}{\sqrt{2}} + \sin \frac{1}{\sqrt{2}} \right)$$

$$= \frac{\pi}{\sqrt{2}} e^{-\frac{1}{\sqrt{2}}} \left(\cos \frac{1}{\sqrt{2}} + \sin \frac{1}{\sqrt{2}} \right)$$

$$\Rightarrow \int_0^{\infty} \frac{\cos x}{1+x^4} dx$$

$$= \frac{1}{2} \cdot 2\pi i (\operatorname{Res}(f, z_1) + \operatorname{Res}(f, z_2))$$

$$= \frac{\pi}{2\sqrt{2}} e^{-\frac{1}{\sqrt{2}}} \left(\cos \frac{1}{\sqrt{2}} + \sin \frac{1}{\sqrt{2}} \right)$$

$$(2) \int_0^{\infty} \frac{\cos x}{(1+x^2)^3} dx$$

与 (1) 同理

$$\int_0^{\infty} \frac{\cos x}{1+x^4} dx$$

$$= \frac{1}{2} \int_{-\infty}^{+\infty} \frac{\cos x}{1+x^4} dx$$

$$= \pi i \cdot \text{Res}(f, i)$$

$$= \pi i \cdot \frac{1}{2!} \frac{d^2}{dz^2} [e^{iz} \cdot (z+i)^{-3}] \Big|_{z=i}$$

$$= \frac{7\pi}{16e}$$

$$(3) \int_{-\infty}^{+\infty} \frac{x \sin x}{x^2 - 2x + 2} dx$$

$$f(z) = \frac{z e^{iz}}{z^2 - 2z + 2}$$

$$\int_{-\infty}^{+\infty} f(z) dz = 2\pi i \cdot \text{Res}(f, 1+i)$$

$$= 2\pi i \cdot \frac{z \cdot e^{iz}}{[z - (1-i)]} \Big|_{z=1+i}$$

$$= \frac{\pi}{e} [\cos| - \sin| + i(\cos| + \sin|)]$$

$$\Rightarrow \int_{-\infty}^{+\infty} \frac{x \sin x}{x^2 - 2x + 2} dx = \frac{\pi}{e} (\cos 1 + \sin 1)$$

$$(4) \int_0^{\infty} \frac{\sin(a+2n)x - \sin ax}{(1+x^2) \sin x} dx, \quad a > -1, n \in \mathbb{Z}$$

$$f(z) = \frac{e^{i(a+2n)z} - e^{iaz}}{(1+z^2) \sin z}$$

$$\int_0^{\infty} \frac{1}{x} dx = \frac{1}{2} \int_{-\infty}^{+\infty} f(z) dz$$

$$= \pi i \cdot \text{Res}[f, z_1] + [f, z_2]$$

$$= \pi \cdot e^{-a-1} \cdot \frac{1-e^{-2n}}{1-e^{-2}}$$

[4]

$$(1) \text{ V.P. } \int_{-\infty}^{+\infty} \frac{dx}{x(x-1)(x-2)}$$

$$= \lim_{\delta \rightarrow 0} \left[\int_{-\infty}^{-\delta} \frac{dx}{x(x-1)(x-2)} + \int_{\delta}^{1-\delta} \frac{dx}{x(x-1)(x-2)} \right]$$

$$+ \int_{1+\delta}^{2-\delta} \frac{dx}{x(x-1)(x-2)} + \int_{2+\delta}^{\infty} \frac{dx}{x(x-1)(x-2)}]$$

$$\frac{1}{x(x-1)(x-2)} = \frac{1}{2(x-2)} - \frac{1}{x-1} + \frac{1}{2x}$$

$$\int_{-\infty}^{-\delta} \frac{dx}{x(x-1)(x-2)} = \lim_{N \rightarrow \infty} \int_{-N}^{-\delta} \left[\frac{1}{2(x-2)} - \frac{1}{x-1} + \frac{1}{2x} \right] dx$$

$$= \lim_{N \rightarrow \infty} \frac{1}{2} \ln \left[\frac{(N+1)^2}{N(N+2)} \right] + \frac{1}{2} \ln \frac{\delta(\delta+2)}{(\delta+1)^2}$$

$$= \frac{1}{2} \ln \frac{\delta(\delta+2)}{(\delta+1)^2}$$

$$\int_{\delta}^{1-\delta} \sim dx = \frac{1}{2} \ln \left[\frac{(1-\delta)^2(1+\delta)}{(2-\delta)\delta^3} \right]$$

$$\int_{1+\delta}^{2-\delta} \sim dx = \frac{1}{2} \ln \left[\frac{\delta}{1-\delta} \cdot \frac{\delta^2}{(1-\delta)^2} \cdot \frac{2-\delta}{1+\delta} \right]$$

$$\int_{2+\delta}^{\infty} \sim dx = \frac{1}{2} \ln \frac{(1+\delta)^2}{\delta(2+\delta)}$$

$$\text{V.P.} \int_{-\infty}^{+\infty} \frac{dx}{x(x-1)(x-2)}$$

$$= \lim_{\delta \rightarrow 0} \frac{1}{2} \ln \left[\frac{\delta(\delta+2)}{(\delta+1)^2} \cdot \frac{(1-\delta)^3(1+\delta)}{(2-\delta)\delta^3} \cdot \frac{\delta^3(2-\delta)}{(1-\delta)^3(1+\delta)} \cdot \frac{(1+\delta)^2}{\delta(2+\delta)} \right]$$

$$= 0$$

$$I(2) \int_0^{\infty} \frac{\sin(x+a) \sin(x-a)}{x^2 - a^2} dx, \quad a > 0$$

$$= -\frac{1}{2} \int_0^{\infty} \frac{\cos(2x) - \cos(2a)}{x^2 - a^2} dx$$

$$= -\frac{1}{4} \int_{-\infty}^{+\infty} \frac{\cos(2x) - \cos(2a)}{x^2 - a^2} dx$$

$$f(z) = \frac{e^{i2z} - e^{i2a}}{z^2 - a^2} \Rightarrow \frac{g(z)}{z - a}$$

$$\int_{C_r(a)} f(z) dz = \int_{\pi}^0 \frac{g(a + re^{i\alpha})}{re^{i\alpha}} \cdot re^{i\alpha} i d\alpha$$

$$= -i \int_0^{\pi} g(a + re^{i\alpha}) d\alpha$$

$$\rightarrow -i\pi g(a) = -i\pi \operatorname{Res}(f, a)$$

$$\int_{-\infty}^{+\infty} f(x) dx = i\pi [\operatorname{Res}(f, a) + \operatorname{Res}(f, -a)]$$

$$= i\pi \left. \frac{e^{i2z} - e^{i2a}}{z+a} \right|_{z=a} + i\pi \left. \frac{e^{i2z} - e^{i2a}}{z-a} \right|_{z=-a}$$

$$= -\frac{\pi}{a} \sin(2a)$$

$$\Rightarrow \int_0^\infty \frac{\sin(x+a)\sin(x-a)}{x^2-a^2} dx = \frac{\pi}{4a} \sin(2a)$$

$$(3) \int_0^\infty \frac{x - \sin x}{x^3(1+x^2)}$$

$$f(z) = \frac{iz - e^{iz}}{z^3(1+z^2)}$$

$$\int_{-R}^{-r} f(z) dz + \int_{C_n} f(z) dz + \int_r^R f(z) dz + \int_{C_R} f(z) dz$$

$$= 2\pi i [\text{Res}(f, i) + \text{Res}(f, -i)]$$

$$\text{Res}(f, i) + \text{Res}(f, -i) = -\frac{e+e^{-1}}{2}$$

$$= -\cosh 1$$

$$\int_{C_n} f(z) dz = -i \int_0^\pi \frac{ine^{ia} - e^{ine^{ia}}}{r^2 \cdot e^{i2a}} da$$

$$\lim_{n \rightarrow 0} \frac{ine^{ia} - e^{ine^{ia}}}{n^2 e^{i2a} (1 + n^2 e^{i2a})} = \frac{1}{2}$$

$$\int_{-\infty}^{+\infty} f(z) dz = 2\pi i \cdot (-\cosh 1) + \frac{\pi}{2} i$$

$$\begin{aligned} \int_0^{\infty} \frac{x - \sin x}{x^3(1+x^2)} dx &= \frac{1}{2} \int_{-\infty}^{+\infty} \frac{x - \sin x}{x^3(1+x^2)} dx \\ &= \frac{\pi}{2} \left(\frac{1}{2} - e - \frac{1}{e} \right) \end{aligned}$$

$$4) \int_{-\infty}^{\infty} \frac{e^{px} - e^{qx}}{1 - e^x} dx, \quad 0 < p < 1, \quad 0 < q < 1$$

$$\frac{\pi}{\alpha} \cot \frac{\pi \beta}{\alpha} = \sum_{n=0}^{\infty} \left\{ \frac{1}{n\alpha + \beta} - \frac{1}{n\alpha + (\alpha - \beta)} \right\}$$

$$\int_{-\infty}^{+\infty} \frac{e^{px} - e^{qx}}{1 - e^x} dx$$

$$= \sum_{n=0}^{\infty} \int_0^{\infty} [e^{-(n+p)x} - e^{-(n+1-p)x}] dx$$

$$- \sum_{n=0}^{\infty} \int_0^{\infty} [e^{-(n+q)x} - e^{-(n+1-q)x}] dx$$

$$= \sum_{n=0}^{\infty} \left(\frac{1}{n+p} - \frac{1}{n+1-p} \right) - \sum_{n=0}^{\infty} \left(\frac{1}{n+q} - \frac{1}{n+1-q} \right)$$

$$= \pi \cot(p\pi) - \pi \cot(q\pi)$$